P2 Chapter 11: Integration

Modelling with Differential Equations

Forming differential equations

Differential equations are useful because regularly in real-life, the rate of change of a variable is based on its current value. For example in Year 1, we saw a property of exponential growth is that the **rate of change is proportional to the current value**:

Q The rate of increase of a rabbit population (with population P, where time is t) is **proportional to** the current population.
Form a differential equation, and find its general solution.

? Form equation

? Solve it

(Notice by the way that $e^{kt} = \left(e^k\right)^t$, and since e^k is a constant, we could always write $y = A \cdot B^x$. i.e. The general solution is 'any generic exponential function', not just restricted to those with e as the base. However, it is customary to write Ae^{kt}).

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Recall from GCSE that k is the 'constant of proportionality'.

$$\frac{dP}{dt} \propto P \quad \therefore \quad \frac{dP}{dt} = kP$$

$$\frac{1}{P}\frac{dP}{dt} = k$$

$$\int \frac{1}{P}\, dP = \int k\, dt$$

$$\ln|P| = kt + C$$

$$P = e^{kt+C}$$

$$P = \boldsymbol{Ae^{kt}} \quad (where\ A = e^{C})$$

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Further Example

[Textbook] Water in a manufacturing plant is held in a large cylindrical tank of diameter 20m. Water flows out of the bottom of the tank through a tap at a rate proportional to the cube root of the volume.

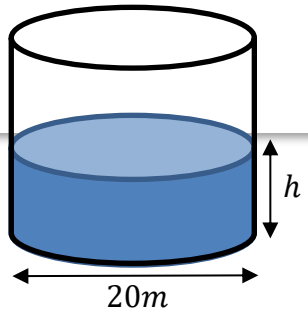
(a) Show that t minutes after the tap is opened, $\frac{dh}{dt} = -k\sqrt[3]{h}$ for some constant k.

(b) Show that the general solution of this differential equation may be written $h = (P - Qt)^{\frac{3}{2}}$, where P and Q are constants.

Initially the height of the water is 27m. 10 minutes later, the height is 8m.

(c) Find the values of the constants P and Q.

(d) Find the time in minutes when the water is at a depth of 1m.



a

?

b

?

c

?

d

?

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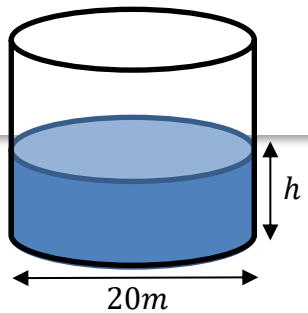

a

$V = \pi r^2 h = 100\pi h$

$\frac{dV}{dh} = 100\pi$

$\frac{dV}{dt} = -c\sqrt[3]{V}$

$= -c\sqrt[3]{100\pi h}$

$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{1}{100\pi} \times \left(-c\sqrt[3]{100\pi h}\right)$

$\therefore \frac{dh}{dt} = -k\sqrt[3]{h}$ where $k = \frac{c\sqrt[3]{100\pi}}{100\pi}$

(a) is a 'connected rate of change' question in the style we saw in Chapter 9. Model the fact we have a cylinder.

If water is flowing out, the rate of change is negative.

Use chain rule.

b

$\frac{dh}{dt} = -kh^{\frac{1}{3}} \rightarrow h^{-\frac{1}{3}}\frac{dh}{dt} = -k$

$\int h^{-\frac{1}{3}} dh = \int -k\, dt$

$\frac{3}{2}h^{\frac{2}{3}} = -kt + c$

$h^{\frac{2}{3}} = -\frac{2}{3}kt + \frac{2}{3}c$

$h = \left(-\frac{2}{3}kt + \frac{2}{3}c\right)^{\frac{3}{2}}$

$h = (P - Qt)^{\frac{3}{2}}$

c

$t = 0, h = 27$

$27 = P^{\frac{3}{2}} \Rightarrow P = 9$

$t = 10, h = 8$

$8 = (9 - 10Q)^{\frac{3}{2}}$

$4 = 9 - 10Q$

$Q = \frac{1}{2}$

d

$h = \left(9 - \frac{1}{2}t\right)^{\frac{3}{2}}$

$1 = \left(9 - \frac{1}{2}t\right)^{\frac{3}{2}}$

$t = 16$ minutes

Test Your Understanding

Edexcel C4 June 2005 Q8

Liquid is pouring into a container at a constant rate of 20 $cm^3\ s^{-1}$ and is leaking out at a rate proportional to the volume of the liquid already in the container.

(*a*) Explain why, at time t seconds, the volume, V cm^3, of liquid in the container satisfies the differential equation

$$\frac{dV}{dt} = 20 - kV,$$

where k is a positive constant.

The container is initially empty.

(*b*) By solving the differential equation, show that

$$V = A + Be^{-kt},$$

giving the values of A and B in terms of k. **(6)**

Given also that $\frac{dV}{dt} = 10$ when $t = 5$,

(*c*) find the volume of liquid in the container at 10 s after the start. **(5)**

(a)

?

Teachers/Students: I recommend also looking at Edexcel Jan 2008 Q8 which has a part (a) similar to the previous example.

(b)

?

:)

?

(6)

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(a)	$\frac{dV}{dt}$ is the rate of increase of volume (with respect to time)	B1
	$-kV$: k is constant of proportionality and the negative shows decrease (or loss)	
	giving $\frac{dV}{dt} = 20 - kV$ ∗ These Bs are to be awarded independently	B1
		(2)

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(b)	$\int \frac{1}{20-kV}\,dV = \int 1\,dt$ separating variables	M1	
	$-\frac{1}{k}\ln(20-kV) = t \quad (+C)$	M1 A1	
	Using $V = 0, t = 0$ to evaluate the constant of integration	M1	
	$c = -\frac{1}{k}\ln 20$		
	$t = \frac{1}{k}\ln\left(\frac{20}{20-kV}\right)$		
	Obtaining answer in the form $V = A + Be^{-kt}$	M1	
	$V = \frac{20}{k} - \frac{20}{k}e^{-kt}$ Accept $\frac{20}{k}(1-e^{-kt})$	A1	(6)

	$\frac{dV}{dt} = 20e^{-kt}$ Can be implied	M1	
	$\frac{dV}{dt} = 10, t = 5 \Rightarrow 10 = 20e^{-kt} \Rightarrow k = \frac{1}{5}\ln 2 \approx 0.139$	M1 A1	
	At $t = 10$, $V = \frac{75}{\ln 2}$ awrt 108	M1 A1	(5)

Exercise 11.11

Pearson Pure Year 2
Pages 99-100

Extension:
[STEP 2011 Q7]

In this question, you may assume that $\ln(1+x) \approx x - \frac{1}{2}x^2$ when $|x|$ is small.

The height of the water in a tank at time t is h. The initial height of the water is H and water flows into the tank at a constant rate. The cross-sectional area of the tank is constant.

(i) Suppose that water leaks out at a rate proportional to the height of the water in the tank, and that when the height reaches $\alpha^2 H$, where α is a constant greater than 1, the height remains constant. Show that

$$\frac{\mathrm{d}h}{\mathrm{d}t} = k(\alpha^2 H - h),$$

for some positive constant k. Deduce that the time T taken for the water to reach height αH is given by

$$kT = \ln\left(1 + \frac{1}{\alpha}\right),$$

and that $kT \approx \alpha^{-1}$ for large values of α.

(ii) Suppose that the rate at which water leaks out of the tank is proportional to $\sqrt{h}$ (instead of h), and that when the height reaches $\alpha^2 H$, where α is a constant greater than 1, the height remains constant. Show that the time T' taken for the water to reach height αH is given by

$$cT' = 2\sqrt{H}\left(1 - \sqrt{\alpha} + \alpha \ln\left(1 + \frac{1}{\sqrt{\alpha}}\right)\right)$$

for some positive constant c, and that $cT' \approx \sqrt{H}$ for large values of α.

Homework Exercise

1 The rate of increase of a population P of rabbits at time t, in years, is given by $\frac{dP}{dt} = kP$, $k > 0$. Initially the population was of size 200.

a Solve the differential equations giving P in terms of k and t. **(3 marks)**

b Given that $k = 3$, find the time taken for the population to reach 4000. **(4 marks)**

c State a limitation of this model for large values of t. **(1 mark)**

2 The mass M at time t of the leaves of a certain plant varies according to the differential equation

$$\frac{dM}{dt} = M - M^2$$

a Given that at time $t = 0$, $M = 0.5$, find an expression for M in terms of t. **(5 marks)**

b Find a value of M when $t = \ln 2$. **(2 marks)**

c Explain what happens to the value of M as t increases. **(1 mark)**

3 The thickness of ice x, in cm, on a pond is increasing at a rate that is inversely proportional to the square of the existing thickness of ice. Initially, the thickness is 1 cm. After 20 days, the thickness is 2 cm.

a Show that the thickness of ice can be modelled by the equation $x = \sqrt[3]{\frac{7}{20}t + 1}$. **(7 marks)**

b Find the time taken for the ice to increase in thickness from 2 cm to 3 cm. **(2 marks)**

4 A mug of tea, with a temperature T °C is made and left to cool in a room with a temperature of 25 °C. The rate at which the tea cools is proportional to the difference in temperature between the tea and the room.

a Show that this process can be described by the differential equation $\frac{dT}{dt} = -k(T - 25)$, explaining why k is a positive constant. **(3 marks)**

Initially the tea is at a temperature of 85 °C. 10 minutes later the tea is at 55 °C.

b Find the temperature, to 1 decimal place, of the tea after 15 minutes. **(7 marks)**

Homework Exercise

5 The rate of change of the surface area of a drop of oil, A mm^2, at time t minutes can be modelled by the equation $\frac{dA}{dt} = \frac{A^{\frac{3}{2}}}{10t^2}$

Given that the surface area of the drop is 1 mm^2 at $t = 1$,

a find an expression for A in terms of t **(7 marks)**

b show that the surface area of the drop cannot exceed $\frac{400}{361}$ mm^2. **(2 marks)**

6 A bath tub is modelled as a cuboid with a base area of 6000 cm^2. Water flows into the bath tub from a tap at a rate of 12 000 cm^3/min. At time t minutes, the depth of water in the bath tub is h cm. Water leaves the bottom of the bath through an open plughole at a rate of $500h$ cm^3/min.

a Show that t minutes after the tap has been opened, $60\frac{dh}{dt} = 120 - 5h$. **(3 marks)**

When $t = 0$, $h = 6$ cm.

b Find the value of t when $h = 10$ cm. **(5 marks)**

7 **a** Express $\frac{1}{P(10\,000 - P)}$ using partial fractions. **(3 marks)**

The deer population, P, in a reservation can be modelled by the differential equation

$$\frac{dP}{dt} = \frac{1}{20\,000}P(10\,000 - P)$$

where t is the time in years since the study began. Given that the initial deer population is 2500,

b solve the differential equation giving your answer in the form $P = \frac{a}{b + ce^{-0.5t}}$ **(6 marks)**

c Find the maximum deer population according to the model. **(2 marks)**

Homework Exercise

8 Liquid is pouring into a container at a constant rate of $40\,\text{cm}^3\,\text{s}^{-1}$ and is leaking from the container at a rate of $\frac{1}{4}V\,\text{cm}^3\,\text{s}^{-1}$, where $V\,\text{cm}^3$ is the volume of liquid in the container.

a Show that $-4\frac{\mathrm{d}V}{\mathrm{d}t} = V - 160$. **(2 marks)**

Given that $V = 5000$ when $t = 0$,

b find the solution to the differential equation in the form $V = a + b\mathrm{e}^{-\frac{1}{4}t}$, where a and b are constants to be found **(7 marks)**

c write down the limiting value of V as $t \to \infty$. **(1 mark)**

9 Fossils are aged using a process called carbon dating. The amount of carbon remaining in a fossil, R, decreases over time, t, measured in years. The rate of decrease of carbon is proportional to the remaining carbon.

a Given that initially the amount of carbon is R_0, show that $R = R_0\mathrm{e}^{-kt}$ **(4 marks)**

It is known that the half-life of carbon is 5730 years. This means that after 5730 years the amount of carbon remaining has reduced by half.

b Find the exact value of k. **(3 marks)**

c A fossil is found with 10% of its expected carbon remaining. Determine the age of the fossil to the nearest year. **(3 marks)**

Homework Answers

1 **a** $y = 200e^{kt}$ **b** 1 year

c The population could not increase in size in this way forever due to limitations such as available food or space.

2 **a** $M = \dfrac{e^t}{1 + e^t}$ **b** $\frac{2}{3}$ **c** M approaches 1

3 **a** $\dfrac{dx}{dt} = \dfrac{k}{x^2} \Rightarrow \frac{1}{3}x^3 = kt + c$

$t = 0, x = 1 \Rightarrow c = \frac{1}{3} \Rightarrow t = 20, x = 2 \Rightarrow k = \frac{7}{60}$

$\frac{1}{3}x^3 = \frac{7}{60}t + \frac{1}{3} \Rightarrow x = \sqrt[3]{\left(\frac{7}{20}t + 1\right)}$

b $x = 3$, $t = 74.3$ days. So it takes 54.3 days to increase from 2 cm to 3 cm.

4 **a** The difference in temperature is $T - 25$. The tea is cooling, so there should be a negative sign. k has to be positive or the tea would be warming.

b 46.2°C

5 **a** $\int A^{-\frac{3}{2}}\,dA = \frac{1}{10}\int t^{-2}dt \Rightarrow \dfrac{-2}{\sqrt{A}} = \dfrac{-1}{10t} + C \Rightarrow C = -\dfrac{19}{10}$

$\Rightarrow \dfrac{-2}{\sqrt{A}} = \dfrac{-1}{10t} - \dfrac{19}{10} \Rightarrow \sqrt{A} = \left(\dfrac{-20t}{-1 - 19t}\right)$

$\Rightarrow A = \left(\dfrac{20t}{1 + 19t}\right)^2$

b As $t \to \infty$, $A \to \left(\dfrac{20}{19}\right)^2 = \dfrac{400}{361}$ from below

6 **a** $V = 6000h \Rightarrow \dfrac{dV}{dh} = 6000, \dfrac{dV}{dt} = 12000 - 500h,$

$\dfrac{dh}{dt} = \dfrac{dV}{dt} \div \dfrac{dV}{dh} = \dfrac{1}{6000}(12000 - 500h)$

$60\dfrac{dh}{dt} = 120 - 5h$

b $t = 12\ln\left(\dfrac{9}{7}\right)$

7 **a** $\dfrac{\left(\dfrac{1}{10000}\right)}{P} + \dfrac{\left(\dfrac{1}{10000}\right)}{10000 - P}$

b $P = \dfrac{10000}{1 + 3e^{-0.5t}}$ so $a = 10000$, $b = 1$ and $c = 3$.

c 10 000 deer

8 **a** $\dfrac{dV}{dt} = 40 - \dfrac{1}{4}V \Rightarrow -4\dfrac{dV}{dt} = V - 160$

b $V = 160 + 4840e^{-\frac{1}{4}t}$, $a = 160$ and $b = 4840$

c $V \to 160$

9 **a** $\dfrac{dR}{dt} = -kR \Rightarrow \displaystyle\int \frac{1}{R}dR = -k\int dt$

$\Rightarrow \ln R = -kt + c \Rightarrow R = e^{-kt+c}$

$\Rightarrow R = Ae^{-kt} \Rightarrow R_0 = Ae^0 \Rightarrow A = R_0 \Rightarrow R = R_0e^{-kt}$

b $k = \dfrac{1}{5730}\ln 2$

c $0.1R_0 = R_0e^{\frac{1}{5730}\ln\left(\frac{1}{2}\right) \times t}$

$\ln(0.1) = \dfrac{1}{5730}\ln\left(\dfrac{1}{2}\right) \times t \Rightarrow t \approx 19035$